ERM

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October 30, 2019

Mr. Doug Langworthy
City of Cadillac WWTP
200 N. Lake Street
Cadillac, Michigan 49601

Reference: 0501867.0106

Subject: Whole Effluent Toxicity Test Results

Dear Doug,

Enclosed please find the final results of the following Chronic Toxicity Tests performed on samples of the City of Cadillac WWTP Outfall 001A effluent.

- 10 October 2019, Chronic *Ceriodaphnia dubia* Toxicity Test
- 8 October 2019, Chronic *Pimephales promelas* Toxicity Test

If you have any questions concerning this report or if I can be of any further assistance to you, please feel free to contact me at (616) 738-7308 or via e-mail at bruce.rabe@erm.com.

Yours sincerely,

Bruce A. Rabe

Bruce A. Rabe
Director, Aquatic Toxicology Laboratory

BAR:km

Enclosures: Whole Effluent Toxicity Test Report

cc: Cindy Tomaszewski
File

FINAL REPORT

Chronic Toxicity Test
Freshwater Invertebrate,
Ceriodaphnia dubia
EPA Test Method 1002.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Sample: Outfall 001A

Testing Period: 10 – 17 October 2019

Laboratory I.D. Number: 101019-9

PA DEP ID No. 68-04227
NJ DEP ID No. MI013

Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424

Test Overview

Permittee:	City of Cadillac WWTP
Location:	1121 Plett Rd. Cadillac, MI 49601
Contact:	Mr. Doug Langworthy
Telephone #:	231.878.4437
NPDES Permit #:	MI0020257
Permit Requirements:	Acute Toxicity = Report Only Chronic Toxicity Limit = 1.1 TUc
Test Sample:	Outfall 001A
Receiving Water:	Clam River
Testing Date:	10 – 17 October 2019
Sample Date(s):	9 October 2019 11 October 2019 14 October 2019
Test/Method:	Daphnid, *Ceriodaphnia dubia*, Survival and Reproduction Test EPA 821-R-02-013 Method 1002.0.
QC Objectives:	Test data met all test acceptability criteria, except where noted below.
Data Qualifiers:	None

DATA SUMMARY

Effluent Concentrations (%)	**Survival (%)**	**Reproduction (Average Young/Female)**
Control	100	36.7
13	100	38.5
25	100	38.6
50	100	40.1
76	90	35.5
100	100	39.7

TEST RESULTS

48-Hour LC_{50}	>100%
NOEC	100%
LOEC	>100%
ChV	>100%
MSDp	18.5%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	0

TEST CONCLUSION

In accordance with the NPDES permit requirement for the City of Cadillac WWTP, this toxicity test did not exhibit either acute or chronic toxicity.

Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0501867.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777

ERM Testing Method

Ceriodaphnia dubia – Survival and Reproduction Toxicity Test

Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (µm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius (°C) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Ceriodaphnia dubia used to initiate this test were obtained from individual, in-house cultures and were less than 24-hours old, and had an age range of 0 to 8 hours at test initiation. Test organisms used to initiate this test were released from adults which met acceptable performance criteria (i.e., ≥15 young/surviving female within 3 broods and obtained from a brood of at least 8 young) and were maintained in reconstituted moderately hard water prior to test initiation.

The *Ceriodaphnia dubia* test was conducted using 30-milliliter (mL) disposable polystyrene containers containing 15 mL of control water or test solution. One *Ceriodaphnia dubia* was added to each test chamber with ten replicate chambers per treatment. Each *Ceriodaphnia dubia* test chamber was fed a 0.2-mL suspension consisting of yeast-Cerophyll-trout chow (YCT) and green algae (*Raphidocelis subcapitata*) mixture daily.

The test solutions were renewed daily during the exposure by transferring the adult daphnid, by way of a wide bore pipette, into fresh control water or test solution.

Percent survival of exposed *Ceriodaphnia dubia* was determined by inspecting for adult mortality daily. Mortality was defined as no body or appendage movement after gentle prodding. Production of young was also determined by daily inspections and enumeration. When 60 percent of the surviving females in the control treatment produced three broods, mean reproduction was determined by calculating the average number of live young produced per female for each treatment.

The test was conducted at a temperature of 25 ± 1°C under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Ceriodaphnia dubia* survival and reproduction. An NOEC is defined as the highest effluent concentration that does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration that does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration. Significant differences in *Ceriodaphnia dubia* survival were determined using the Fisher's Exact Test.

Prior to the determination of any significant differences in *Ceriodaphnia dubia* reproduction, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or reproduction) and converted to a chronic toxic unit (TUc) for reporting purposes by 100/ChV, unless the LOEC was greater than 100 percent in which case the TUc was reported as 0.

To evaluate acute toxicity, a 48-hour LC_{50} and corresponding 95 percent confidence interval was also calculated, where possible. The LC_{50} value estimate was determined by using one of the following statistical methods: graphical, Spearman-Karber, Trimmed Spearman-Karber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 48-hour LC_{50} value was converted to an acute toxic unit (TUa) by $100/LC_{50}$.

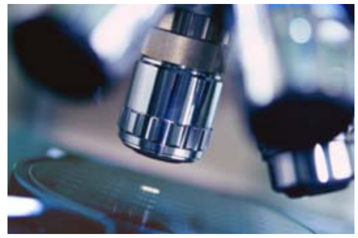
For 48-hour LC_{50} values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 percent to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 percent to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.9.4.3 software program.

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms and the precision of the testing procedure. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC_{25}) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC_{25} of a particular reference toxicant test does not fall within the expected range of ± two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM’S STANDARD TEST METHOD

Test was performed in accordance with ERM’s standard test method (see page 3).

Appendix A

Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

Environmental Resources Management

Ceriodaphnia dubia - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation

Permittee/Client:	City of Cadillac WWTP		Control/Dilution Water:	RMHW
Effluent/Location:	Outfall 001A		Organism Batch #:	187-19
Lab I.D.#:	101019-9		Organism Age:	8-16 hrs.
Beginning Date:	10/10/19	Time: 1400	QC Review:	KM
Ending Date:	10/17/19	Time: 1300	QC Review Date:	10/28/19

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	101019-9	101119-14	101519-8	116-19	119-19	-
Temperature	° C	1	1	1	--	--	--
Dissolved Oxygen	mg / L	11.8	10.2	9.8	--	--	--
pH	S.U.	7.1	7.0	7.1	7.8	7.9	-
Conductivity	umhos/cm	919	945	843	322	316	-
Alkalinity	mg / L $CaCO_3$	82	72	92	62	58	-
Hardness	mg / L $CaCO_3$	220	200	160	82	80	-
Total Ammonia	mg / L NH_3	<0.01	0.10	0.01	--	--	--
Total Residual Chlorine	mg / L Cl_2	<0.01	<0.01	<0.01	<0.01	<0.01	-
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	SPR	MS	SPR	KM	KM	-

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)		Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	1200		0	KM	1	116-19
13%	156	1044		1	MS	1	116-19
25%	300	900		2	SAR	2	116-19
50%	600	600		3	DM	2	116-19
76%	912	288		4	KM	2	~~116-19~~ 119-19 KM 10/15
100%	1200	0		5	KM	3	119-19
				6	KM	3	119-19
				7	KM	-	-

Environmental Resources Management

Ceriodaphnia dubia - Chronic Toxicity Test
Water Quality Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 101019-9

Water Quality Data:

Dissolved Oxygen (mg/L)

Day														
Meter #	3	3	3	5	3	3	5	5	5	5	3	3	3	3
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.4	7.1	8.2	7.6	8.1	7.6	8.2	8.1	8.3	7.9	8.3	8.0	8.2	8.1
13%	8.4	7.1	8.3	7.5	8.1	7.6	8.1	8.2	8.4	7.9	8.3	8.0	8.2	8.1
25%	8.5	6.9	8.2	7.5	8.2	7.7	8.2	8.2	8.4	7.9	8.3	8.0	8.3	8.1
50%	8.5	6.9	8.2	7.7	8.3	7.7	8.2	8.2	8.4	7.8	8.4	8.0	8.3	8.0
76%	8.5	6.8	8.2	7.5	8.5	7.7	8.3	8.2	8.4	7.6	8.4	7.9	8.3	8.1
100%	8.4	6.4	8.3	7.9	8.6	7.6	8.4	8.2	8.4	7.7	8.5	7.9	8.4	8.1

pH (S.U.)

Day														
Meter #	10	10	10	9	10	10	9	8	9	9	10	10	10	10
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.9	7.7	8.0	7.7	8.1	7.8	8.0	7.7	7.9	7.6	7.9	7.7	7.9	7.7
13%	--	7.7	--	7.8	--	7.7	--	7.8	--	7.7	--	7.7	--	7.8
25%	--	7.7	--	7.8	--	7.8	--	7.8	--	7.7	--	7.8	--	7.8
50%	--	7.8	--	7.9	--	7.9	--	7.9	--	7.8	--	7.8	--	7.8
76%	--	7.8	--	8.0	--	8.0	--	8.0	--	7.6	--	7.9	--	7.9
100%	7.5	7.8	7.5	8.0	7.4	8.1	7.4	8.0	7.3	7.9	7.4	8.0	7.3	8.0

Conductivity (umhos / cm)

Day														
Meter #	4	--	4	--	4	--	3	--	3	--	4	--	4	--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	322	--	321	--	322	--	317	--	323	--	317	--	315	--
13%	397	--	395	--	400	--	409	--	405	--	378	--	378	--
25%	465	--	466	--	470	--	477	--	479	--	441	--	442	--
50%	607	--	607	--	622	--	626	--	628	--	565	--	565	--
76%	770	--	771	--	786	--	782	--	782	--	693	--	693	--
100%	910	--	908	--	927	--	912	--	922	--	809	--	809	--
		--		--		--		--		--		--		--

Temperature (° C)

Day														
Meter #	3	3	3	5	3	3	5	5	5	5	3	3	3	3
Treatment	0	1		2		3 MM		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	24	24	24	25	24	25	24	24	24	24	24	24	24
13%	24	24	24	24	25	24	25	24	24	24	24	24	24	24
25%	24	24	24	24	25	24	25	24	24	24	24	24	24	24
50%	24	24	24	24	25	24	25	24	24	24	24	24	25	24
76%	24	24	25	24	25	24	25	24	24	24	24	24	23	24
100%	24	24	25	24	25	24	25	24	24	24	24	24	23	24
			~~24~~											

10/11 MM I = Initial Chemistry F = Final Chemistry

Note: D.O. meter also used for temperature measurement unless otherwise noted.

Environmental Resources Management

Ceriodaphnia dubia - Chronic Toxicity Test Survival and Reproduction Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 101019-9

Treatment (% Effluent)	Day No.	Replicate 1	2	3	4	5	6	7	8	9	10	Average Young/ Female	Number of Live Adults (% Sur.)	Average Young/ Female % CV
Control	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	—	—	—	—	—	—	—	—	—	—		10	
	4	6	8	7	5	6	4	7	6	5	4		10	
	5	7	14	14	14	15	10	14	13	13	—		10	
	6	—	—	—	—	17	—	19	—	—	12		10	
	7	20	19	26	16	—	13	—	18	17	18		10	
Totals:		33	41	47	35	38	27	40	37	35	34	36.7	(100)	14.6
# Broods (% 3rd Brood)		3	3	3	3	3	3	3	3	3	3	(100)		
13%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	—	—	—	—	—	—	—	—	—	—		10	
	4	4	8	6	5	5	6	6	6	7	6		10	
	5	14	17	13	12	13	15	14	12	20	9		10	
	6	—	—	—	—	21	20	16	—	—	—		10	
	7	21	20	24	17	—	—	—	14	16	18		10	
Totals:		39	45	43	34	39	41	36	32	43	33	38.5	(100)	11.9
25%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	—	—	—	—	—	—	—	—	—	—		10	
	4	6	8	6	5	6	5	5	6	6	6		10	
	5	15	9	15	18	14	13	13	11	15	10		10	
	6	—	21	22	—	21	20	17	—	17	—		10	
	7	19	—	—	23	—	—	— 35	16	—	18		10	
Totals:		40	38	43	46	41	38	30	33	38	34	38.6	(100)	10.5
50%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	—	—	—	—	—	—	—	—	—	—		10	
	4	6	7	6	6	5	6	6	5	5	7		10	
	5	14	16	12	12	16	13	15	12	14	14		10	
	6	—	—	—	22	21	20	20	—	—	—		10	
	7	19	24	18	—	—	—	—	20	21	19		10	
Totals:		39	47	36	40	42	39	41	37	40	40	40.1	(100)	7.5
76%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	—	—	—	—	—	—	—	—	—	—		10	
	4	X	6	6	5	6	6	6(1)	—	5	9		9	
	5	X	17	16	15	13	12	16	9	10	12		9	
	6	X	—	—	23	19	19	20	17	—	—		9	
	7	X	23	21	—	—	—	—	14	13	17		9	
Totals:		0	46	43	43	38	37	42	40	28	38	35.5	(90)	37.8
100%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	—	—	—	—	—	—	—	—	—	—		10	
	4	7	7	7	6	4	8	8	2	6	9		10	
	5	15	12	12	14	11	13	14	—	12	12		10	
	6	—	20	—	21	19	22	18	12	—	—		10	
	7	21	—	22	—	—	—	—	24	19	20		10	
Totals:		43	39	41	41	34	43	40	38	37	41	39.7	(100)	7.0

X = DEAD ADULT 1X = DEAD ADULT, ONE YOUNG PRODUCED BEFORE DEATH -- = NO YOUNG RECORDED
(E) = ABORTED EMBRYOS /EGGS (1) = ONE DEAD YOUNG (S) = SPLIT BROOD * = 4th BROOD EXCLUDED FROM TOTAL

Environmental Resources Management

Ceriodaphnia dubia **- Chronic Toxicity Test Support Data**

Table 3
Page 2 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 101019-9

Brood Board Information:

Replicate	1	2	3	4	5	6	7	8	9	10	Brood Board Date: 09/30/19
Chamber Number	10	11	13	25	30	40	50	53	55	59	Young Age Range: 8-16 hours

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
YCT Batch #:	20-19	21-19	21-19	21-19	21-19	21-19	21-19	→
Algae Batch #:	21-19	22-19	22-19	22-19	22-19	22-19	22-19	—
Observation Time:	1400	1430	1430	1030	1430	1400	1300	1300
Initials:	MS	AM	AM	MS	MS	SOR	KM	KM
Date:	10/10/19	10/11/19	10/12/19	10/13/19	10/14/19	10/15/19	10/16/19	10/17/19

Comment Section:

Day	Date	Initials	Comments
2	10/12/19	AM	Many organisms in the higher concentrations (50 & 100%) were stuck in a filamentous material and appeared to be struggling to free themselves

CETIS Analytical Report

Report Date:	19 Oct-19 05:48 (p 1 of 2)		
Test Code/ID:	36B9CC3B / 09-1814-6107		

Ceriodaphnia 7-d Survival and Reproduction Test ERM

Analysis ID:	16-9880-0347	**Endpoint:**	7d Survival Rate	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:47	**Analysis:**	STP 2xK Contingency Tables	**Status Level:**	1

Batch ID:	04-2253-2827	**Test Type:**	Reproduction-Survival (7d)	**Analyst:**	Lab Tech
Start Date:	10 Oct-19 14:00	**Protocol:**	EPA/821/R-02-013 (2002)	**Diluent:**	Reconstituted Water
Ending Date:	17 Oct-19 13:00	**Species:**	Ceriodaphnia dubia	**Brine:**	
Test Length:	6d 23h	**Taxon:**	Branchiopoda	**Source:**	In-House Culture **Age:** <24

Sample ID:	13-5655-6162	**Code:**	50DB6782	**Project:**	WET Compliance Testing
Sample Date:	09 Oct-19 07:00	**Material:**	POTW Effluent	**Source:**	City of Cadillac WWTP
Receipt Date:	10 Oct-19 10:00	**CAS (PC):**		**Station:**	Outfall 001A
Sample Age:	31h (1 °C)	**Client:**	City of Cadillac WWTP		

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU
Untransformed	C > T	100	>100	n/a	1

Fisher Exact/Bonferroni-Holm Test

Control	vs	Group	Test Stat	P-Type	P-Value	Decision(α:5%)
Lab Water		13	1.0000	Exact	1.0000	Non-Significant Effect
		25	1.0000	Exact	1.0000	Non-Significant Effect
		50	1.0000	Exact	1.0000	Non-Significant Effect
		76	0.5000	Exact	1.0000	Non-Significant Effect
		100	1.0000	Exact	1.0000	Non-Significant Effect

Test Acceptability Criteria

		TAC Limits			
Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	1	0.8	>>	Yes	Passes Criteria

Data Summary

Conc-%	Code	NR	R	NR + R	Prop NR	Prop R	%Effect
0	L	10	0	10	1	0	0.0%
13		10	0	10	1	0	0.0%
25		10	0	10	1	0	0.0%
50		10	0	10	1	0	0.0%
76		9	1	10	0.9	0.1	10.0%
100		10	0	10	1	0	0.0%

7d Survival Rate Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
13		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
25		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
50		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
76		0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
100		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

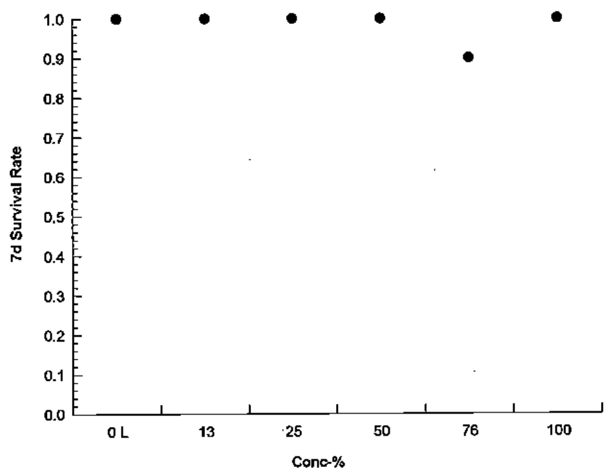
7d Survival Rate Binomials

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
13		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
25		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
50		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
76		0/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
100		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1

Ceriodaphnia 7-d Survival and Reproduction Test **ERM**

Analysis ID:	16-9880-0347	**Endpoint:**	7d Survival Rate	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:47	**Analysis:**	STP 2xK Contingency Tables	**Status Level:**	1

Graphics



Analyst: [illegible] QA: [illegible]

CETIS Analytical Report

Report Date:	19 Oct-19 05:48 (p 1 of 2)
Test Code/ID:	36B9CC3B / 09-1814-6107

Ceriodaphnia 7-d Survival and Reproduction Test — ERM

Analysis ID:	05-4069-6388	**Endpoint:**	Reproduction	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:48	**Analysis:**	Nonparametric-Control vs Treatments	**Status Level:**	1

Batch ID:	04-2253-2827	**Test Type:**	Reproduction-Survival (7d)	**Analyst:**	Lab Tech
Start Date:	10 Oct-19 14:00	**Protocol:**	EPA/821/R-02-013 (2002)	**Diluent:**	Reconstituted Water
Ending Date:	17 Oct-19 13:00	**Species:**	Ceriodaphnia dubia	**Brine:**	
Test Length:	6d 23h	**Taxon:**	Branchiopoda	**Source:**	In-House Culture — **Age:** <24

Sample ID:	13-5655-6162	**Code:**	50DB6782	**Project:**	WET Compliance Testing
Sample Date:	09 Oct-19 07:00	**Material:**	POTW Effluent	**Source:**	City of Cadillac WWTP
Receipt Date:	10 Oct-19 10:00	**CAS (PC):**		**Station:**	Outfall 001A
Sample Age:	31h (1 °C)	**Client:**	City of Cadillac WWTP		

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	PMSD
Untransformed	C > T	100	>100	n/a	1.0	18.45%

Per MDEGLE

Steel Many-One Rank Sum Test

Control	vs	Conc-%	Test Stat	Critical	Ties	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	114.5	75	3	18	Asymp	0.9664	Non-Significant Effect
		25	116.5	75	6	18	Asymp	0.9780	Non-Significant Effect
		50	129	75	4	18	Asymp	0.9992	Non-Significant Effect
		76	118	75	3	18	Asymp	0.9843	Non-Significant Effect
		100	127.5	75	5	18	Asymp	0.9988	Non-Significant Effect

Test Acceptability Criteria

Attribute	Test Stat	TAC Limits Lower	Upper	Overlap	Decision
Control Resp	36.7	15	>>	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	156.483	31.2967	5	0.7154	0.6146	Non-Significant Effect
Error	2362.5	43.75	54			
Total	2518.98		59			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Bartlett Equality of Variance Test	34.77	15.09	1.7E-06	Unequal Variances
Distribution	Shapiro-Wilk W Normality Test	0.7727	0.9459	3.0E-08	Non-Normal Distribution

Reproduction Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	10	36.7	32.87	40.53	36	27	47	1.693	14.59%	0.00%
13		10	38.5	35.23	41.77	39	32	45	1.447	11.89%	-4.90%
25		10	38.6	35.7	41.5	38	33	46	1.284	10.52%	-5.18%
50		10	40.1	37.96	42.24	40	36	47	0.9481	7.48%	-9.26%
76		10	35.5	25.91	45.09	39	0	46	4.238	37.75%	3.27%
100		10	39.7	37.7	41.7	40.5	34	43	0.8825	7.03%	-8.17%

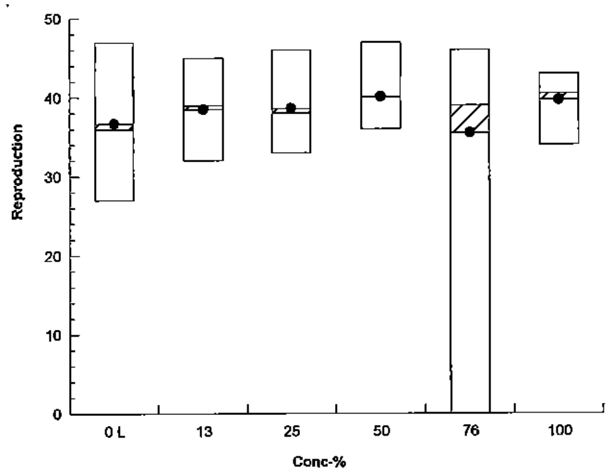
Reproduction Detail

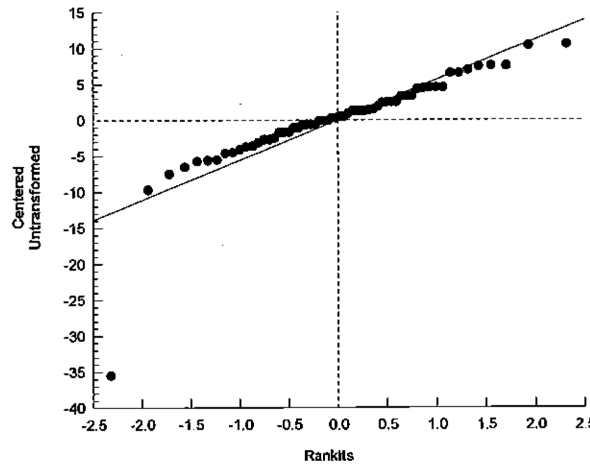
Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	33	41	47	35	38	27	40	37	35	34
13		39	45	43	34	39	41	36	32	43	33
25		40	38	43	46	41	38	35	33	38	34
50		39	47	36	40	42	39	41	37	40	40
76		0	46	43	43	38	37	42	40	28	38
100		43	39	41	41	34	43	40	38	37	41

Ceriodaphnia 7-d Survival and Reproduction Test — **ERM**

Analysis ID:	05-4069-6388	**Endpoint:**	Reproduction	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:48	**Analysis:**	Nonparametric-Control vs Treatments	**Status Level:**	1

Graphics





ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	DS
ADDRESS:	1121 Plett Rd Cadillac MI	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Terl.	10-8-19 10-9-19	7AM 7AM	C	2 - 1 gal	pH= 7.24 s.u. NH_3= .158 mg/L	101019-9	Temp. 1 (°C) ☐ On Ice	D.O. 11.6 mg/L	pH 7.1 s.u.	Cond 919 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]	Test Material:	Test Type:	Test Species:
	X Water/Wastewater ___ Sediment ___ Product	X Acute X Chronic ___ Other	___ *Ceriodaphnia dubia* ___ *Daphnia magna* ___ *Daphnia pulex* ___ Fathead minnow (*Pimephales promelas*) ___ Rainbow Trout (*Oncorhynchus mykiss*) ___ Sheepshead minnow (*Cyprinodon variegatus*) ___ Silverside minnow (*Menidia beryllina*) ___ *Americamysis bahia* ___ *Hyalella azteca* ___ *Chironomus dilutus* ___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
[signature] / City of Cadillac	10-9-19	2pm	Steven P. [signature] / ERM	10/10/19	1000

* See Instructions for Sample Collection on Back of Sheet

February 2018

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ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	DS
ADDRESS:	1121 Plett Rd. Cadillac MI	PHONE NUMBER:	231 -775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Test	10-10-19 10-11-19	7AM 7AM	C	3-1gal	pH= 7.09 s.u. NH₃= .253. mg/L	101219-14	Temp. 1 (°C) ☐ On Ice	D.O. 10.2 mg/L	pH 7.0 s.u.	Cond 945 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
X Water/Wastewater
__ Sediment
__ Product

Test Type:
X Acute
X Chronic
__ Other

Test Species:
__ *Ceriodaphnia dubia*
__ *Daphnia magna*
__ *Daphnia pulex*
__ Fathead minnow (*Pimephales promelas*)
__ Rainbow Trout (*Oncorhynchus mykiss*)
__ Sheepshead minnow (*Cyprinodon variegatus*)
__ Silverside minnow (*Menidia beryllina*)
__ *Americamysis bahia*
__ *Hyalella azteca*
__ *Chironomus dilutus*
__ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
[signature] / City of Cadillac	10-11-19	2PM	[signature] / ERM	10/12/19	1100

* See Instructions for Sample Collection on Back of Sheet

February 2018

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ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	DS
ADDRESS:	1121 Plett Road Cadillac MI	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
TERT.	10-13-19 10-14-19	7AM 7AM	C	1 gal	pH= 7.15 s.u. NH_3= <0.3 mg/L	101514-8	Temp. 6 (°C) ☒ On Ice	D.O. 9.8 mg/L	pH 8.71 s.u.	Cond 813 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
X Water/Wastewater
___ Sediment
___ Product

Test Type:
X Acute
X Chronic
___ Other

Test Species:
___ *Ceriodaphnia dubia*
___ *Daphnia magna*
___ *Daphnia pulex*
___ Fathead minnow (*Pimephales promelas*)
___ Rainbow Trout (*Oncorhynchus mykiss*)
___ Sheepshead minnow (*Cyprinodon variegatus*)
___ Silverside minnow (*Menidia beryllina*)
___ *Americamysis bahia*
___ *Hyalella azteca*
___ *Chironomus dilutus*
___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
[signature]	10/14/19	2pm	[signature]	10/15/19	1130

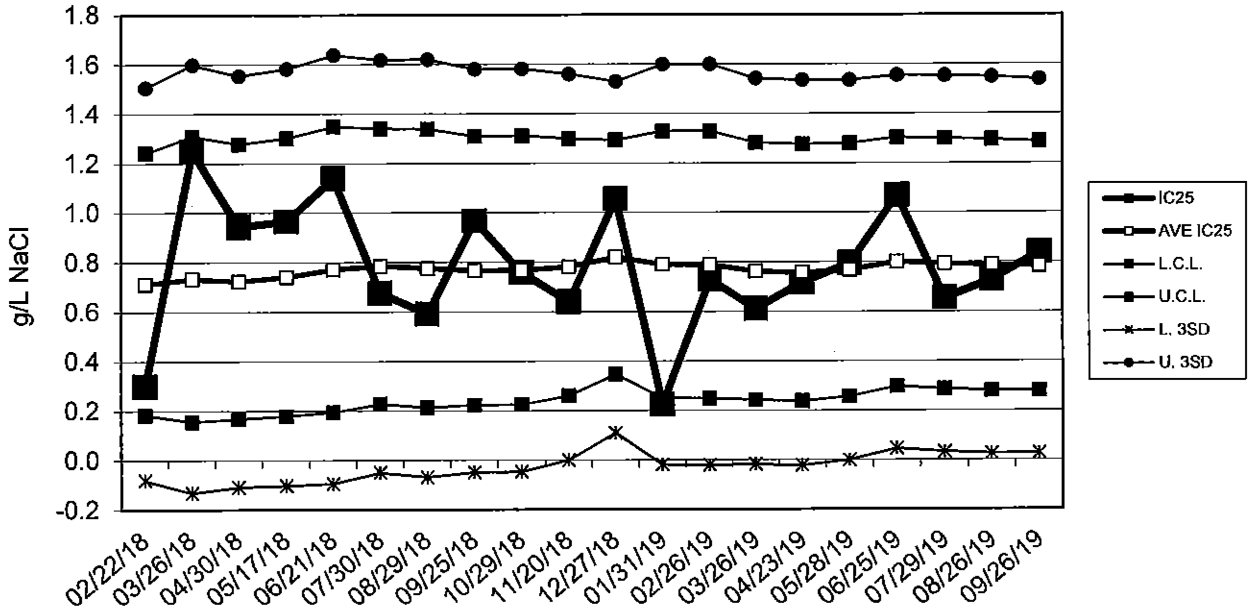
* See Instructions for Sample Collection on Back of Sheet

February 2018

ERM

Environmental Resources Management

Standard Reference Toxicant Data



Chronic *Ceriodaphnia dubia* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT Lower	Upper	Survival (%)	CONTROL Reproduction (ave. young)	CV (%)
02/22/18	0.30	0.7	0.18	1.24	100	17.8	35.0
03/26/18	1.25	0.7	0.16	1.31	90	32.5	38.5
04/30/18	0.94	0.7	0.17	1.28	100	32.0	25.5
05/17/18	0.97	0.7	0.18	1.30	100	30.0	38.6
06/21/18	1.14	0.8	0.19	1.35	80	35.2	8.2
07/30/18	0.68	0.8	0.23	1.34	100	25.5	16.3
08/29/18	0.59	0.8	0.21	1.34	100	30.1	26.2
09/25/18	0.97	0.8	0.22	1.31	100	27.6	26.7
10/29/18	0.76	0.8	0.22	1.31	100	32.7	24.8
11/20/18	0.64	0.8	0.26	1.30	100	34.8	15.2
12/27/18	1.06	0.8	0.35	1.29	100	26.8	43.7
01/31/19	0.23	0.8	0.25	1.33	100	34.7	14.9
02/26/19	0.73	0.8	0.25	1.33	100	27.9	9.3
03/26/19	0.61	0.8	0.24	1.28	100	40.2	9.9
04/23/19	0.72	0.8	0.24	1.28	100	36.1	25.4
05/28/19	0.79	0.8	0.26	1.28	100	37.6	3.1
06/25/19	1.07	0.8	0.30	1.30	100	29.4	26.7
07/29/19	0.65	0.8	0.29	1.30	100	33.7	14.6
08/26/19	0.73	0.8	0.28	1.29	100	30.4	23.5
09/26/19	0.84	0.8	0.28	1.29	100	24.8	27.4

Appendix B
MDEGLE Reporting Form

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than 1 test.

1. NAME OF FACILITY (on NPDES permit) City of Cadillac WWTP	2. NPDES PERMIT # M I 0 0 2 0 2 5 7

1. RECEIVING WATER (as designated in permit) Clam River	2. OUTFALL 001A	5. RECEIVING WATER CONCENTRATION (if known)

6. TEST LAB (Name and Address) Environmental Resources Management
3352 128th Avenue
Holland, MI 49424-9263

7. TEST START DATE 10/10/19	8. TEST END DATE 10/17/19	9. AGE RANGE OF ORGANISMS AT TEST START 8 – 16 Hours Old	10. REPORT DATE 10/30/19

11. NAME OF PERSON CONDUCTING TEST Lab Contact: Mark Baker	12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Bruce Rabe (616) 738 – 7308

13. SAMPLE COLLECTION DATES	14. DATE RECEIVED	15. ARRIVAL TEMP (C°)
Sample 1: 10/09/19	Sample 1: 10/10/19	Sample 1: 1
Sample 2: 10/11/19	Sample 2: 10/12/19	Sample 2: 1
Sample 3: 10/14/19	Sample 3: 10/15/19	Sample 3: 1

16. DATE OF FIRST USE	17. TOTAL RESIDUAL CHLORINE (mg/l)	18. AMMONIA (mg/l as N)
Sample 1: 10/10/19	Sample 1: <0.01	Sample 1: <0.01
Sample 2: 10/12/19	Sample 2: <0.01	Sample 2: 0.10
Sample 3: 10/15/19	Sample 3: <0.01	Sample 3: 0.01

19. WAS SAMPLE DECHLORINATED?	20. DESCRIBE DECHLORINATION (if any)
Sample 1: ☐ YES ☒ NO IF YES ➡	
Sample 2: ☐ YES ☒ NO IF YES ➡	
Sample 3: ☐ YES ☒ NO IF YES ➡	

21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION

☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE

22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time.) N/A

23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡	24. STATE MESH SIZE OF FILTER

25. EFFLUENT SAMPLE TYPE (check one type for each sample)			26. IDENTIFY THE DILUENT (O_1) CONTROL ____RMHW__
Sample 1: ☒ 24-HR COMPOSITE	☐ GRAB/COMPOSITE (give # of grabs)____	☐ GRAB SAMPLE	IDENTIFY THE SECONDARY (O_2) CONTROL (if used)
Sample 2: ☒ 24-HR COMPOSITE	☐ GRAB/COMPOSITE (give # of grabs)____	☐ GRAB SAMPLE	__________________
Sample 3: ☒ 24-HR COMPOSITE	☐ GRAB/COMPOSITE (give # of grabs)____	☐ GRAB SAMPLE	

27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTION

CONCENTRATION OF EFFLUENT (%)	O_1	O_2	13%	25%	50%	76%	100%
48-HOUR SURVIVAL (%)	100	N/A	100	100	100	100	100
7-DAY MEAN REPRODUCTION/FEMALE	36.7	N/A	38.5	38.6	40.1	35.5	39.7
7-DAY MEAN SURVIVAL (%)	100	N/A	100	100	100	90	100

28. 48-HOUR LC_{50} (%) >100	29. TU_a (acute toxic units) 0

30. 7-DAY CHRONIC VALUE (%) >100	31. NOEC 100%	32. LOEC >100%	33. TU_c (chronic toxic units) 0

EQP 5819 (02/10)

FINAL REPORT

Chronic Toxicity Test
Freshwater Vertebrate,
Pimephales promelas
EPA Test Method 1000.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Sample: Outfall 001A

Testing Period: 8 – 15 October 2019

Laboratory I.D. Number: 100819-16

PA DEP ID No. 68-04227
NJ DEP ID No. MI013

Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424

Test Overview

Permittee:	City of Cadillac WWTP
Location:	1121 Plett Rd. Cadillac, MI 49601
Contact:	Mr. Doug Langworthy
Telephone #:	231.878.4437
NPDES Permit #:	MI0020257
Permit Requirements:	Report
Test Sample:	Outfall 001A
Receiving Water:	Clam River
Testing Date:	8 – 15 October 2019
Sample Date(s):	7 October 2019 9 October 2019 11 October 2019
Test/Method:	Fathead Minnow, *Pimephales promelas*, Survival and Growth Test EPA 821-R-02-013 Method 1000.0.
QC Objectives:	Test data met all test acceptability criteria, except where noted below.
Data Qualifiers:	None

DATA SUMMARY

Effluent Concentrations (%)	**Survival (%)**	**Growth Average Wt./ Organism (mg)**
Control	100	0.603
13	92.5	0.577
25	95	0.606
50	100	0.588
76	90	0.547
100	80*	0.536*

* Statistically lower than the control (P=0.05)

TEST RESULTS

96-Hour LC_{50}	>100%
NOEC	76%
LOEC	100%
ChV	87.2%
MSDp (Growth)	27.1%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	1.1

TEST CONCLUSION

In accordance with the NPDES permit requirements for the City of Cadillac WWTP, this toxicity test did not exhibit acute toxicity but did exhibit chronic toxicity.

Bruce A. Rabe

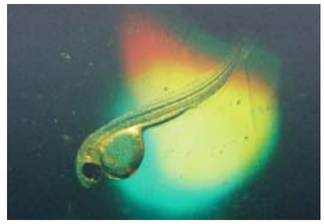
Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0501867.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777

ERM.

ERM Testing Method

Pimephales promelas – Survival and Growth Toxicity Test



Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (µm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius (°C) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Pimephales promelas used to initiate this test were obtained from in-house cultures and were less than 24-hours old at test initiation. Test organisms were maintained in reconstituted moderately hard water prior to test initiation.

The *Pimephales promelas* test was conducted using 300 to 500-milliliter (mL) disposable polypropylene containers containing 250 mL of control water or test solution. Ten fish were randomly added to each test chamber with four replicate chambers per treatment. Each *Pimephales promelas* test chamber was fed 0.1 mL of a concentrated suspension of less than 24-hour old live brine shrimp nauplii (*Artemia* sp.) two times per day. Test solutions were renewed daily during the exposure by replacing approximately 90 percent of the 24-hour old solution with fresh control water or appropriate test solution. Prior to test solution renewal, uneaten and dead brine shrimp, along with other debris, were removed from the bottom of the test chambers.

Percent survival of exposed *Pimephales promelas* was determined daily by enumeration of live organisms. Mortality was defined as no body movement after gentle prodding. At the termination of the chronic test, larvae in each test chamber were counted, dried, and weighed to the nearest 0.01 milligram (mg) on an analytical balance.

The test was conducted at a temperature of 25 ± 1°C under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Pimephales promelas* survival and growth. An NOEC is defined as the highest effluent concentration which does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration which does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration.

Prior to the determination of any significant differences in *Pimephales promelas* survival and growth, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed, followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test.
Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or growth) and converted to a chronic toxic unit (TUc) for reporting purposes by 100/ChV, unless the LOEC was greater than 100 percent in which case the TUc was reported as 0.

To evaluate acute toxicity, a 96-hour LC_{50} and corresponding 95 percent confidence interval were also calculated, where possible. The LC_{50} value estimate was determined by using one of the following statistical methods: graphical, Spearman-Karber, Trimmed Spearman-Karber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 96-hour LC_{50} value was converted to an acute toxic unit (TUa) by $100/LC_{50}$.

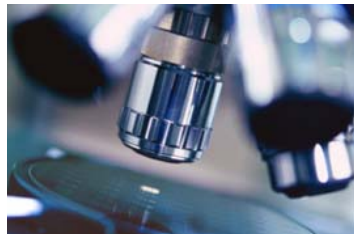
For 96-hour LC_{50} values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in the 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.9.4.3 software program

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC_{25}) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC_{25} of a particular reference toxicant test does not fall within the expected range of $\pm$ two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM'S STANDARD TEST METHOD

Test was performed in accordance with ERM's standard test method (see page 3).

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

Environmental Resources Management

Pimephales promelas - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation

Permittee/Client:	City of Cadillac WWTP	Control/Dilution Water:	RMHW
Effluent/Location:	Outfall 001A	Organism Batch #:	177-19
Lab I.D.#:	100819-16	Organism Age:	<24 hrs old
Beginning Date:	10/08/19 Time: 1700	QC Review:	KM
Ending Date:	10/15/19 Time: 1630	QC Review Date:	10/20/19

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	100819-16	101019-9	101219-14	116-19	119-19	—
Temperature	° C	|	|	|	--	--	--
Dissolved Oxygen	mg / L	10.9	11.8	10.2	--	--	--
pH	S.U.	7.0	7.1	7.0	7.8	— 7.9	—
Conductivity	umhos/cm	928	919	945	322	— 316	—
Alkalinity	mg / L $CaCO_3$	76	62	72	62	— 58	—
Hardness	mg / L $CaCO_3$	240	220	200	82	— 80	—
Total Ammonia	mg / L NH_3	0.02	<0.01	0.10	--	--	--
Total Residual Chlorine	mg / L Cl_2	<0.01	<0.01	<0.01	<0.01	— <0.01	—
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	KM	SPR	MS	KM	— SPR	—

Test Solution Preparation: Test Solution Prepared For Both Species.

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)		Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	1200		0	KM	1	116-19
13%	156	1044		1	Km	1	116-19
25%	300	900		2	KM	2	116-19
50%	600	600		3	MS	2	116-19
76%	912	288		4	SAR	3	116-19
100%	1200	0		5	AM	3	116-19
				6	KM	3	~~116-19~~ 119-19 SPR 10/15/19
				7	SPR	--	--

Environmental Resources Management

Pimephales promelas - Chronic Toxicity Test
Water Quality Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 100819-16

Water Quality Data:

Dissolved Oxygen (mg/L)

	Day													
Meter #	3	5	3	5	3	5	3	3	3	3	5	5	5	3
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.4	6.8	8.4	6.0	8.4	5.9	8.2	5.5	8.1	5.9	8.2	6.1	8.3	7.0
13%	8.4	6.8	8.4	5.9	8.4	6.1	8.3	5.6	8.1	5.7	8.1	5.7	8.4	6.6
25%	8.4	6.5	8.4	5.7	8.5	6.2	8.2	5.7	8.2	5.2	8.2	5.6	8.4	6.5
50%	8.4	6.8	8.4	6.0	8.5	6.1	8.2	5.4	8.3	5.2	8.2	5.7	8.4	6.0
76%	8.4	7.1	8.5	6.0	8.5	6.0	8.2	4.8	8.5	5.3	8.3	5.7	8.4	6.6
100%	8.4	6.6	8.3	5.6	8.4	5.6	8.3	5.2	8.6	5.3	8.4	4.9	8.4	6.1

pH (S.U.)

	Day													
Meter #	10	8	10	10	10	9	10	10	10	10	9	9	9	10
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.8	7.3	8.0	7.3	7.9	7.0	8.0	7.3	8.1	7.3	8.0	7.2	7.9	7.6
13%	--	7.4	--	7.4	--	7.0	--	7.4	--	7.3	--	7.2	--	7.5
25%	--	7.4	--	7.3	--	7.0	--	7.4	--	7.4	--	7.2	--	7.5
50%	--	7.5	--	7.3	--	7.0	--	7.5	--	7.4	--	7.3	--	7.4
76%	--	7.4	--	7.2	--	7.1	--	7.5	--	7.5	--	7.3	--	7.4
100%	7.4	7.3	7.5	7.2	7.5	7.2	7.5	7.6	7.4	7.6	7.4	7.3	7.3	7.6

Conductivity (umhos / cm)

	Day													
Meter #	3	--	4	--	4	--	4	--	4	--	3	--	3	--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	323	--	320	--	322	--	321	--	322	--	317	--	323	--
13%	402	--	393	--	397	--	395	--	400	--	409	--	405	--
25%	469	--	458	--	465	--	466	--	470	--	477	--	479	--
50%	616	--	601	--	607	--	607	--	622	--	626	--	628	--
76%	761	--	763	--	770	--	771	--	786	--	782	--	782	--
100%	898	--	900	--	910	--	908	--	927	--	912	--	922	--

Temperature (° C)

	Day													
Meter #	3	5	3	5	3	5	3	3	3	3	5	5	5	3
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	25	24	25	24	25	24	24	25	24	25	24	24	25
13%	24	25	24	25	24	25	24	24	25	24	25	24	24	25
25%	24	25	24	25	24	25	24	24	25	24	25	24	24	25
50%	24	25	24	25	24	26	24	24	25	24	25	24	24	25
76%	24	25	24	25	24	25	25	24	25	24	25	24	24	25
100%	24	25	24	25	24	25	25	24	25	24	25	24	24	25

I = Initial Chemistry F = Final Chemistry

Note: D.O. meter also used for temperature measurement unless otherwise noted.

Environmental Resources Management

Pimephales promelas - Chronic Toxicity Test Survival Data

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 100819-16

Survival Data:

Treatment (% Effluent)	Rep.	# Live Organisms Day 0	1	2	3	4	5	6	7	Rep.	# Live Organisms Day 0	1	2	3	4	5	6	7	96 Hour Survival Summary Total Live Initial	Final	% Survival
Control	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100
13%	A	10	10	9	9	9	9	9	9	B	10	10	10	10	10	10	10	10	40	39	97.5
25%	A	10	10	10	10	10	9	9	9	B	10	10	10	10	10	10	10	10	40	39	97.6
50%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100
76%	A	10	10	10	10	10	10	10	10	B	10	10	9	9	9	9	9	6	40	39	97.5
100%	A	10	9	89	9	9	8	7	6	B	10	8 9	8 9	9	9	8	8	8	40	36	90
			89 ms 10/10									m 10/10									

Treatment (% Effluent)	Rep.	# Live Organisms (ms 10/10) Day 0	1	2	3	4	5	6	7	Rep.	# Live Organisms (SPR 10/11/19) Day 0	1	2	3	4	5	6	7	7 Day Survival Summary Total Live Initial	Final	% Survival
Control	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100
13%	C	10	10	10	10	10	10	9	9	D	10	10	10	10	10	10	10	10	40	37	92.5
25%	C	10	10	10	10	10	10	10	10	D	10	10	9	9	9	9	9	9	40	38	95
50%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100
76%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	36	90
100%	C	10	9	9	9	9	9	9	9	D	10	89	9	9	9	9	9	9	40	32	80
												ms 10/10									

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Time:	1700	1600	1600	1530	1530	1130	1500	1630
Initials:	AM	AM	ms	SPR	ms	ms	SPR	SPR
Date:	10/8/19	10/9/19	10/10/19	10/11/19	10/12/19	10/13/19	10/14/19	10/15/19

Feeding:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Batch #:	280-19	281-19	282-19	283-19	284-19	285-19	286-19	--
Initials AM:	--	SPR	ms	SPR	ms	ms	ms	--
Initials PM:	ms	ms	BM	ms	ms	ms	ms	--

Oven:

Date In	Time In	Initials		Date Out	Time Out	Initials
10/15/19	1700	SPR		10/16/19	1700	SPR

Comment Section:

Day	Date	Initials	Comments

Environmental Resources Management

Pimephales promelas - Chronic Toxicity Test
Growth Data

Table 3
Page 2 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 100819-16

Pan #	Conc. (% Effluent)	Replicate	Final Weight (mg)	Initial Weight (mg)	Larvae Weight (mg)	# of Initial Organisms	Avg. Wt./ Organism/ Replicate (mg)	Avg. Wt./ Organism/ Treatment (mg)	Avg. Wt./ Organism/ Treatment % CV
Date			10/17/19	10/15/19					
Analyst			sb/km	sb/km					
1	Control	A	19.24	13.44	5.80	10	0.580		
2	Control	B	17.52	11.77	5.75	10	0.575		
3	Control	C	23.09	17.25	5.84	10	0.584		
4	Control	D	19.54	12.83	6.71	10	0.671	0.603	7.6
5	13%	A	21.25	16.37	4.88	10	0.488		
6	13%	B	18.31	11.94	6.37	10	0.637		
7	13%	C	16.45	11.44	5.01	10	0.501		
8	13%	D	18.90	12.10	6.80	10	0.680	0.577	16.7
9	25%	A	19.70	14.23	5.47	10	0.547		
10	25%	B	15.95	10.36	5.59	10	0.559		
11	25%	C	20.40	13.76	6.64	10	0.664		
12	25%	D	20.30	13.77	6.53	10	0.653	0.606	10.1
13	50%	A	21.16	15.05	6.11	10	0.611		
14	50%	B	21.49	14.93	6.56	10	0.656		
15	50%	C	19.50	15.17	4.33	10	0.433		
16	50%	D	19.66	13.16	6.50	10	0.650	0.588	17.9
17	76%	A	19.66	14.19	5.47	10	0.547		
18	76%	B	16.35	12.95	3.40	10	0.340		
19	76%	C	20.12	13.88	6.24	10	0.624		
20	76%	D	18.28	11.53	6.75	10	0.675	0.547	27.0
21	100%	A	17.80	13.47	4.33	10	0.433		
22	100%	B	18.25	13.30	4.95	10	0.495		
23	100%	C	20.99	14.69	6.30	10	0.630		
24	100%	D	17.95	12.08	5.87	10	0.587	0.536	16.6

Quality Assurance

					Final Wt. (mg)
25	Blank	A	18.81	18.85	-0.04
26	Blank	B	14.9	14.91	-0.01

CETIS Analytical Report

Report Date:	19 Oct-19 05:53 (p 1 of 2)
Test Code/ID:	7F9C02C6 / 21-4093-0758

Fathead Minnow 7-d Larval Survival and Growth Test — **ERM**

Analysis ID:	11-7533-7084	**Endpoint:**	7d Survival Rate	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:53	**Analysis:**	Nonparametric-Control vs Treatments	**Status Level:**	1

Batch ID:	06-7560-3831	**Test Type:**	Growth-Survival (7d)	**Analyst:**	Lab Tech
Start Date:	08 Oct-19 17:00	**Protocol:**	EPA/821/R-02-013 (2002)	**Diluent:**	Reconstituted Water
Ending Date:	15 Oct-19 16:30	**Species:**	Pimephales promelas	**Brine:**	
Test Length:	6d 23h	**Taxon:**	Actinopterygii	**Source:**	In-House Culture **Age:** <24

Sample ID:	07-8048-7751	**Code:**	2E854C47	**Project:**	WET Compliance Testing
Sample Date:	07 Oct-19 07:00	**Material:**	POTW Effluent	**Source:**	City of Cadillac WWTP
Receipt Date:	08 Oct-19 13:30	**CAS (PC):**		**Station:**	Outfall 001A
Sample Age:	34h (1 °C)	**Client:**	City of Cadillac WWTP		

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	PMSD
Angular (Corrected)	C > T	76	100	87.18	~~1.316~~ 1.1 per MDEGLG	15.77%

Steel Many-One Rank Sum Test

Control	vs	Conc-%	Test Stat	Critical	Ties	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	14	10	1	6	Asymp	0.3451	Non-Significant Effect
		25	14	10	1	6	Asymp	0.3451	Non-Significant Effect
		50	18	10	1	6	Asymp	0.8333	Non-Significant Effect
		76	16	10	1	6	Asymp	0.6105	Non-Significant Effect
		100*	10	10	0	6	Asymp	0.0417	Significant Effect

Test Acceptability Criteria

Attribute	Test Stat	TAC Limits Lower	TAC Limits Upper	Overlap	Decision
Control Resp	1	0.8	>>	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	0.22941	0.0458819	5	2.134	0.1079	Non-Significant Effect
Error	0.386962	0.0214979	18			
Total	0.616372		23			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Levene Equality of Variance Test	4.961	4.248	0.0050	Unequal Variances
Variances	Mod Levene Equality of Variance Test	1.048	4.248	0.4206	Equal Variances
Distribution	Shapiro-Wilk W Normality Test	0.8384	0.884	0.0013	Non-Normal Distribution

7d Survival Rate Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.00%	0.00%
13		4	0.9250	0.7727	1.0000	0.9500	0.8000	1.0000	0.0479	10.35%	7.50%
25		4	0.9500	0.8581	1.0000	0.9500	0.9000	1.0000	0.0289	6.08%	5.00%
50		4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.00%	0.00%
76		4	0.9000	0.5818	1.0000	1.0000	0.6000	1.0000	0.1000	22.22%	10.00%
100		4	0.8000	0.5750	1.0000	0.8500	0.6000	0.9000	0.0707	17.68%	20.00%

Angular (Corrected) Transformed Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	1.412	1.412	1.412	1.412	1.412	1.412	0	0.00%	0.00%
13		4	1.295	1.061	1.529	1.331	1.107	1.412	0.07348	11.35%	8.28%
25		4	1.331	1.181	1.48	1.331	1.249	1.412	0.04705	7.07%	5.77%
50		4	1.412	1.412	1.412	1.412	1.412	1.412	0	0.00%	0.00%
76		4	1.281	0.8621	1.699	1.412	0.8861	1.412	0.1315	20.54%	9.31%
100		4	1.123	0.8501	1.396	1.178	0.8861	1.249	0.08571	15.27%	20.48%

Fathead Minnow 7-d Larval Survival and Growth Test ERM

Analysis ID:	11-7533-7084	**Endpoint:**	7d Survival Rate	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:53	**Analysis:**	Nonparametric-Control vs Treatments	**Status Level:**	1

7d Survival Rate Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	1.0000	1.0000	1.0000	1.0000
13		0.8000	1.0000	0.9000	1.0000
25		0.9000	1.0000	1.0000	0.9000
50		1.0000	1.0000	1.0000	1.0000
76		1.0000	0.6000	1.0000	1.0000
100		0.6000	0.8000	0.9000	0.9000

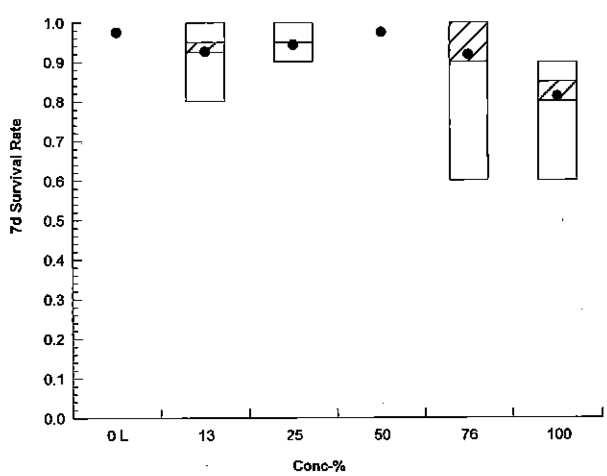
Angular (Corrected) Transformed Detail

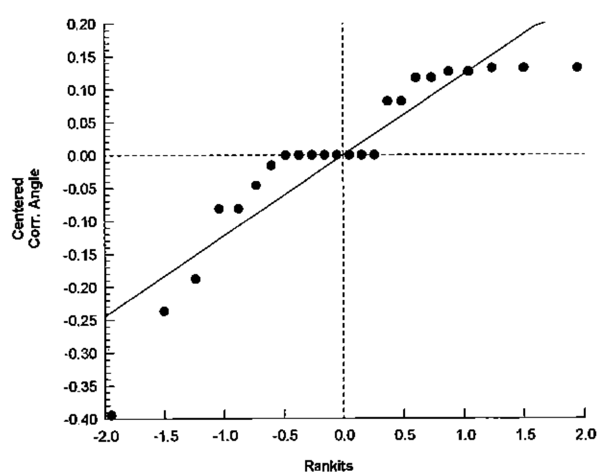
Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	1.412	1.412	1.412	1.412
13		1.107	1.412	1.249	1.412
25		1.249	1.412	1.412	1.249
50		1.412	1.412	1.412	1.412
76		1.412	0.8861	1.412	1.412
100		0.8861	1.107	1.249	1.249

7d Survival Rate Binomials

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	10/10	10/10	10/10	10/10
13		8/10	10/10	9/10	10/10
25		9/10	10/10	10/10	9/10
50		10/10	10/10	10/10	10/10
76		10/10	6/10	10/10	10/10
100		6/10	8/10	9/10	9/10

Graphics





CETIS Analytical Report

Report Date:	19 Oct-19 05:54 (p 1 of 2)
Test Code/ID:	7F9C02C6 / 21-4093-0758

Fathead Minnow 7-d Larval Survival and Growth Test — ERM

Analysis ID:	18-7831-6901	**Endpoint:**	Mean Dry Biomass-mg	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:53	**Analysis:**	Parametric-Control vs Treatments	**Status Level:**	1
Batch ID:	06-7560-3831	**Test Type:**	Growth-Survival (7d)	**Analyst:**	Lab Tech
Start Date:	08 Oct-19 17:00	**Protocol:**	EPA/821/R-02-013 (2002)	**Diluent:**	Reconstituted Water
Ending Date:	15 Oct-19 16:30	**Species:**	Pimephales promelas	**Brine:**	
Test Length:	6d 23h	**Taxon:**	Actinopterygii	**Source:**	In-House Culture **Age:** <24
Sample ID:	07-8048-7751	**Code:**	2E854C47	**Project:**	WET Compliance Testing
Sample Date:	07 Oct-19 07:00	**Material:**	POTW Effluent	**Source:**	City of Cadillac WWTP
Receipt Date:	08 Oct-19 13:30	**CAS (PC):**		**Station:**	Outfall 001A
Sample Age:	34h (1 °C)	**Client:**	City of Cadillac WWTP		

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	PMSD
Untransformed	C > T	76	>76	n/a	1.316	27.06%

Dunnett Multiple Comparison Test

Control vs	Conc-%	Test Stat	Critical	MSD	DF	P-Type	P-Value	Decision(α:5%)
Lab Water	13	0.3757	2.356	0.163	6	CDF	0.6582	Non-Significant Effect
	25	-0.04697	2.356	0.163	6	CDF	0.8148	Non-Significant Effect
	50	0.2168	2.356	0.163	6	CDF	0.7227	Non-Significant Effect
	76	0.8093	2.356	0.163	6	CDF	0.4659	Non-Significant Effect

Test Acceptability Criteria

Attribute	Test Stat	TAC Limits Lower	Upper	Overlap	Decision
Control Resp	0.6025	0.25	>>	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	0.009159	0.0022898	4	0.2391	0.9118	Non-Significant Effect
Error	0.143647	0.0095765	15			
Total	0.152806		19			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Bartlett Equality of Variance Test	4.037	13.28	0.4010	Equal Variances
Distribution	Shapiro-Wilk W Normality Test	0.9421	0.866	0.2625	Normal Distribution

Mean Dry Biomass-mg Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	0.6025	0.5296	0.6754	0.582	0.575	0.671	0.02291	7.60%	0.00%
13		4	0.5765	0.423	0.73	0.569	0.488	0.68	0.04822	16.73%	4.32%
25		4	0.6057	0.5083	0.7032	0.606	0.547	0.664	0.03064	10.12%	-0.54%
50		4	0.5875	0.4206	0.7544	0.6305	0.433	0.656	0.05246	17.86%	2.49%
76		4	0.5465	0.312	0.781	0.5855	0.34	0.675	0.07369	26.97%	9.29%

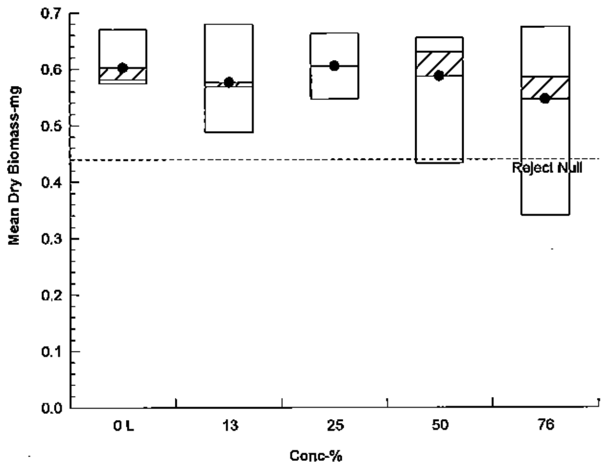
Mean Dry Biomass-mg Detail

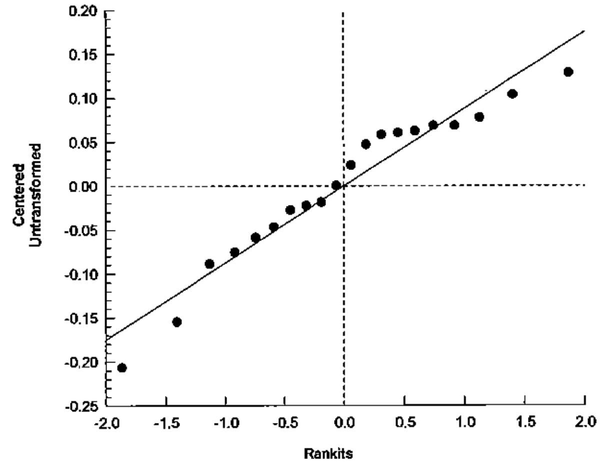
Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	0.58	0.575	0.584	0.671
13		0.488	0.637	0.501	0.68
25		0.547	0.559	0.664	0.653
50		0.611	0.656	0.433	0.65
76		0.547	0.34	0.624	0.675

Fathead Minnow 7-d Larval Survival and Growth Test **ERM**

Analysis ID:	18-7831-6901	**Endpoint:**	Mean Dry Biomass-mg	**CETIS Version:**	CETISv1.9.4
Analyzed:	19 Oct-19 5:53	**Analysis:**	Parametric-Control vs Treatments	**Status Level:**	1

Graphics





ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	DS
ADDRESS:	1121 Plett Rd. Cadillac MI	PHONE NUMBER:	231 775 - 2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM) Temp.	D.O.	pH	Cond
Tert.	10-6-19 10-7-19	7AM 7AM	C	2 - 1gal.	pH= 7.0 s.u. NH3= ~~cor~~ <0.3 mg/L	100819-16	1 (°C) ☒ On Ice	10.9 mg/L	7.0 s.u.	928 umhos/cm
					pH= s.u. NH3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
X Water/Wastewater
___ Sediment
___ Product

Test Type:
X Acute
X Chronic
___ Other

Test Species:
X *Ceriodaphnia dubia*
___ *Daphnia magna*
___ *Daphnia pulex*
X Fathead minnow (*Pimephales promelas*)
___ Rainbow Trout (*Oncorhynchus mykiss*)
___ Sheepshead minnow (*Cyprinodon variegatus*)
___ Silverside minnow (*Menidia beryllina*)
___ *Americamysis bahia*
___ *Hyalella azteca*
___ *Chironomus dilutus*
___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
Dan [signature] / City of Cadillac	10-7-19	2pm	Karla Martin / ERM	10/8/19	1330

* See Instructions for Sample Collection on Back of Sheet

February 2018

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ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	DS
ADDRESS:	1121 Plett Rd Cadillac MI	PHONE NUMBER:	231-775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert.	10-8-19 / 10-9-19	7AM / 7AM	C	2 - 1 gal	pH= 7.24 s.u. NH_3= .158 mg/L	101019-9	Temp. 1 (°C) ☐ On Ice	D.O. 11.6 mg/L	pH 7.1 s.u.	Cond 919 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
X Water/Wastewater
___ Sediment
___ Product

Test Type:
X Acute
X Chronic
___ Other

Test Species:
___ *Ceriodaphnia dubia*
___ *Daphnia magna*
___ *Daphnia pulex*
___ Fathead minnow (*Pimephales promelas*)
___ Rainbow Trout (*Oncorhynchus mykiss*)
___ Sheepshead minnow (*Cyprinodon variegatus*)
___ Silverside minnow (*Menidia beryllina*)
___ *Americamysis bahia*
___ *Hyalella azteca*
___ *Chironomus dilutus*
___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature /Organization	DATE	TIME
[signature] / City of Cadillac	10-9-19	2PM	[signature] / ERM	10/10/19	1000

* See Instructions for Sample Collection on Back of Sheet

February 2018

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ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263
Phone: 616- 399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac	SAMPLER	DS
ADDRESS:	1121 Plett Rd. Cadillac MI	PHONE NUMBER:	231 -775-2841

SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM) Temp. (°C)	D.O.	pH	Cond
Test	10-10-19 10-11-19	7AM 7AM	C	3-1gal	pH= 7.09 s.u. NH_3= .253 mg/L	101219-14	Temp. 1 (°C) ☐ On Ice	D.O. 10.2 mg/L	pH 7.0 s.u.	Cond 945 umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH_3= mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]

Test Material:
- X Water/Wastewater
- ___ Sediment
- ___ Product

Test Type:
- X Acute
- X Chronic
- ___ Other

Test Species:
- ___ *Ceriodaphnia dubia*
- ___ *Daphnia magna*
- ___ *Daphnia pulex*
- ___ Fathead minnow (*Pimephales promelas*)
- ___ Rainbow Trout (*Oncorhynchus mykiss*)
- ___ Sheepshead minnow (*Cyprinodon variegatus*)
- ___ Silverside minnow (*Menidia beryllina*)
- ___ *Americamysis bahia*
- ___ *Hyalella azteca*
- ___ *Chironomus dilutus*
- ___ Other (write in comments section)

COMMENT SECTION:

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
[signature] / City of Cadillac	10-11-19	2PM	[signature] / ERM	10/12/19	1100

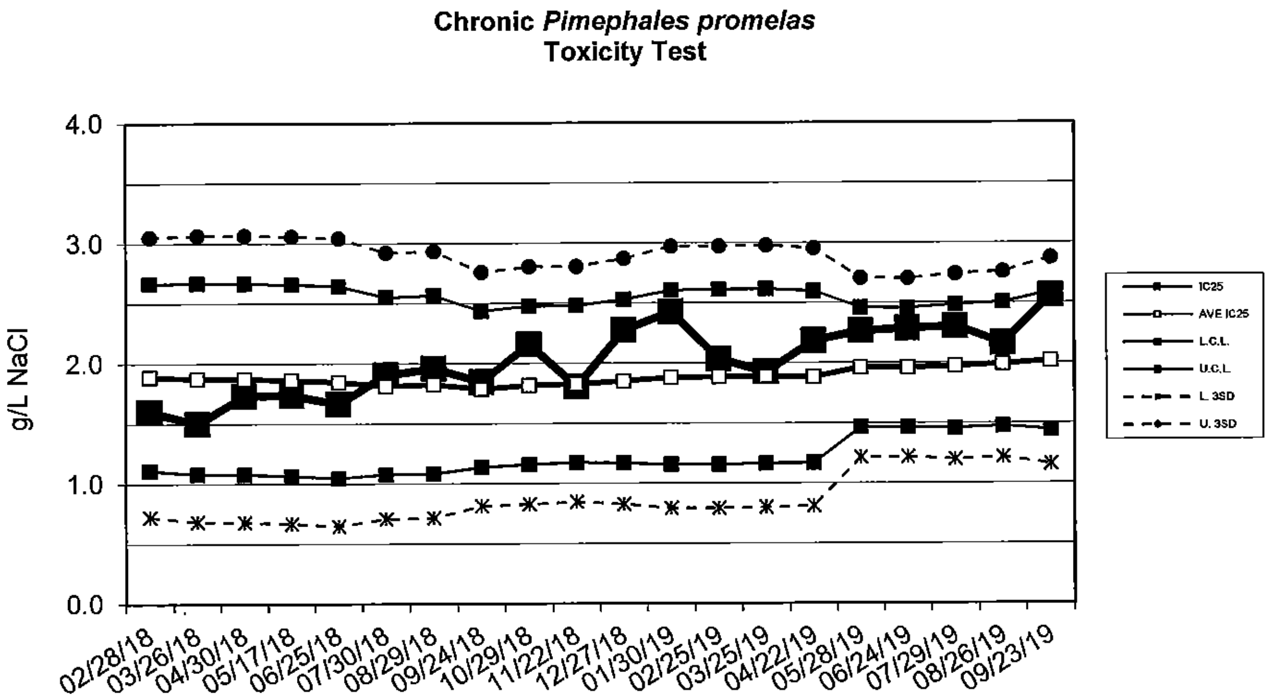
* See Instructions for Sample Collection on Back of Sheet

February 2018

Environmental Resources Management

ERM

Standard Reference Toxicant Data



Chronic *Pimephales promelas* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT Lower	CONTROL LIMIT Upper	Survival (%)	CONTROL Growth (mg)	CV (%)
02/28/18	1.6	1.9	1.1	2.7	92.5	0.44	10.7
03/26/18	1.5	1.9	1.1	2.7	97.5	0.47	3.5
04/30/18	1.7	1.9	1.1	2.7	95	0.45	11.4
05/17/18	1.7	1.9	1.1	2.7	100	0.54	10.8
06/25/18	1.7	1.8	1.0	2.6	95	0.56	17.8
07/30/18	1.9	1.8	1.1	2.6	97.5	0.43	4.3
08/29/18	2.0	1.8	1.1	2.6	100	0.58	9.4
09/24/18	1.8	1.8	1.1	2.4	97.5	0.46	8.2
10/29/18	2.2	1.8	1.2	2.5	97.5	0.45	7.7
11/22/18	1.8	1.8	1.2	2.5	95	0.65	5.2
12/27/18	2.3	1.8	1.2	2.5	97.5	0.64	7.4
01/30/19	2.4	1.9	1.2	2.6	100	0.53	10.5
02/25/19	2.0	1.9	1.2	2.6	95	0.53	10.2
03/25/19	1.9	1.9	1.2	2.6	97.5	0.63	6.0
04/22/19	2.2	1.9	1.2	2.6	100	0.57	2.0
05/28/19	2.3	2.0	1.5	2.5	100	0.68	10.4
06/24/19	2.3	2.0	1.5	2.5	92.5	0.48	11.0
07/29/19	2.3	2.0	1.5	2.5	100	0.51	5.6
08/26/19	2.2	2.0	1.5	2.5	100	0.38	15.0
09/23/19	2.6	2.0	1.4	2.6	97.5	0.52	5.8

Appendix B
MDEGLE Reporting Form

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

FATHEAD MINNOW CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than one test.

1. NAME OF FACILITY (on NPDES permit): City of Cadillac WWTP

2. NPDES PERMIT #: M I 0 0 2 0 2 5 7

1. RECEIVING WATER (as designated in permit): Clam River

2. OUTFALL: 001A

5. RECEIVING WATER CONCENTRATION (if known):

6. TEST LAB (Name and Address): Environmental Resources Management, 3352 128th Avenue, Holland, MI 49424-9263

7. TEST START DATE: 10/08/19

8. TEST END DATE: 10/15/19

9. AGE RANGE OF ORGANISMS AT TEST START: <24 Hrs Old

10. REPORT DATE: 10/30/19

11. NAME OF PERSON CONDUCTING TEST: Lab Contact: Mark Baker

12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT: Bruce Rabe (616) 738 – 7308

13. SAMPLE COLLECTION DATES	14. DATE RECEIVED	15. ARRIVAL TEMP (C°)
Sample 1: 10/07/19	Sample 1: 10/08/19	Sample 1: 1
Sample 2: 10/09/19	Sample 2: 10/10/19	Sample 2: 1
Sample 3: 10/11/19	Sample 3: 10/12/19	Sample 3: 1

16. DATE OF FIRST USE	17. TOTAL RESIDUAL CHLORINE (mg/l)	18. AMMONIA (mg/l as N)
Sample 1: 10/08/19	Sample 1: <0.01	Sample 1: 0.02
Sample 2: 10/10/19	Sample 2: <0.01	Sample 2: <0.01
Sample 3: 10/12/19	Sample 3: <0.01	Sample 3: 0.10

19. WAS SAMPLE DECHLORINATED?

Sample 1: ☐ YES ☒ NO IF YES

Sample 2: ☐ YES ☒ NO IF YES

Sample 3: ☐ YES ☒ NO IF YES

20. DESCRIBE DECHLORINATION (if any)

21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE

22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time. Attach a separate sheet if necessary.) *N/A*

23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES

24. STATE MESH SIZE OF FILTER (if filtered)

25. EFFLUENT SAMPLE TYPE (check one type for each sample)

Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE

Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE

Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs)____ ☐ GRAB SAMPLE

26. IDENTIFY THE DILUENT (O_1) CONTROL: RMHW

IDENTIFY THE SECONDARY (O_2) CONTROL (if used): ________

27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND GROWTH

CONCENTRATION OF EFFLUENT (%)	O_1 (dilution H_2O)	O_2	13%	25%	50%	76%	100%
96-HOUR SURVIVAL (%)	100	N/A	97.5	97.5	100	97.5	90
7-DAY MEAN BIOMASS (mg/initial fish)	0.603	N/A	0.577	0.606	0.588	0.547	0.536
7-DAY MEAN SURVIVAL (%)	100	N/A	92.5	95	100	90	80

28. 96-HOUR LC_{50} (%): >100

29. TU_a (acute toxic units): 0

30. 7-DAY CHRONIC VALUE (%): 87.2

31. NOEC: 76%

32. LOEC: 100%

33. TU_c (chronic toxic units): 1.1

EQP 5820 (02/10)